

Junyeob Baek

Ph.D. Student @ KAIST

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Last updated: Sep. 2026

Research Interest

My research goal is to build “*An embodied machine that can develop intelligence throughout its lifetime, just like humans.*” Specifically, the machine should establish and grow its own lens of worlds, goals, and values in open-ended environments — with interpretability and alignment as guiding concerns.

My current path toward this vision is to build **compositional world models** — understanding the world as reusable structure — and the **recursive reasoning** that runs over them to generalize beyond experience.

I pursue the following research questions, with **Q1** as my current focus throughout the PhD:

- **Q1. How can AI understand the world more systematically and compositionally? – Systematic World Understanding & Generalization (current focus)**
- *Q2. How can AI continuously accumulate and apply knowledge/skills? – Continual Learning*
- *Q3. Can AI develop its own goals following intrinsic motivation? – Autotelic Agents*
- *Q4. How can AI maintain autonomy while sharing human values? – Human-Value Alignment*

Education

Mar. 2025 – Present	Korea Advanced Institute of Science and Technology(KAIST) Ph.D. in Computer Science - Advisor: Sungjin Ahn	Daejeon, Korea
Mar. 2023 – Feb. 2025	Korea Advanced Institute of Science and Technology(KAIST) M.S. in Artificial Intelligence - Advisor: Sungjin Ahn • Thesis: Dreamweaver: Learning Compositional World Models from Pixels	Daejeon, Korea
Mar. 2014 – Feb. 2018	Hanyang University(ERICA) B.S. in Robot Engineering - GPA - 4.29 / 4.5 • <i>Summa Cum Laude</i> • <i>Capstone Project1: Disaster and Rescue Robot Simulation Modeling(Darrsm) - 2nd Prize in College</i> • <i>Capstone Project2: Walk Yourself, Toddler!(WalkYTo) - 3rd Prize in Department</i>	Ansan, Korea

Research Experience

Jun. 2025 – Aug. 2025	Visiting Researcher , Mila-Quebec AI Institute	Montreal, Canada
	• Collaborating with <i>Yoshua Bengio</i>’s research group and <i>LawZero</i> team members • Conducting research project on Mechanistic Interpretability and Chain-of-Thought Faithfulness in LLMs • Main Collaborators: <i>Joumana Ghosn, Pierre-Luc St-Charles, Marc-Antoine Rondeau, Minsu Kim</i>	

Honors & Awards

2026	Best Paper Award , ICML 2026 Workshop on Compositional Learning
2026	Oral Presentation (Top 0.7%, 168/23918) , ICML 2026
2026	Silver Reviewer , ICML 2026
2025	Outstanding Teaching Assistant Award , KAIST

Publications

*Equal contribution

Conference Papers

- C5 Junyeob Baek***, Mingyu Jo*, Minsu Kim, Mengye Ren, Yoshua Bengio, and Sungjin Ahn. “Generative Recursive Reasoning”. In: *Conference on Neural Information Processing Systems (NeurIPS)* (2026).
- C4 Doojin Baek***, **Junyeob Baek***, Mingyu Jo, Hosung Lee, Taegu Kang, Gyubin Lee, and Sungjin Ahn. “BabyTheorist: A Benchmark for Learning to Theorize the World from Observation Alone”. In: *Conference on Neural Information Processing Systems (NeurIPS)* (2026).

- C3 Doojin Baek*, Gyubin Lee*, **Junyeob Baek**, Hosung Lee, and Sungjin Ahn. “Learning to Theorize the World from Observation”. In: *International Conference on Machine Learning (ICML)* (2026). **Oral Presentation (Top 0.7%) • Best Paper Award**, ICML 2026 Workshop on Compositional Learning.
- C2 **Junyeob Baek**, Yi-Fu Wu, Gautam Singh, and Sungjin Ahn. “Dreamweaver: Learning Compositional World Models from Pixels”. In: *International Conference on Learning Representation (ICLR)* (2025).
- C1 Chang Chen, **Junyeob Baek**, Fei Deng, Kenji Kawaguchi, Caglar Gulcehre, and Sungjin Ahn. “PlanDQ: Hierarchical Plan Orchestration via D-Conductor and Q-Performer”. In: *International Conference on Machine Learning (ICML)* (2024).

Workshop Papers

- W1 **Junyeob Baek**, Hosung Lee, Christopher Hoang, Mengye Ren, and Sungjin Ahn. “Discrete JEPA: Learning Discrete Token Representations without Reconstruction”. In: *International Conference on Machine Learning (ICML) Tokenization Workshop* (2025).

Preprints

- P1 Donghwan Chi, Hyomin Kim, Yoojin Oh, Yongjin Kim, Donghoon Lee, Daejin Jo, Jongmin Kim, **Junyeob Baek**, Sungjin Ahn, and Sungwoong Kim. “Slot-MLLM: Object-Centric Visual Tokenization for Multimodal LLM”. *arXiv preprint arXiv:2505.17726* (2025).

Thesis

Junyeob Baek. “Dreamweaver: Learning Compositional World Models from Pixels”. Master’s Thesis, KAIST, Advisor: Sungjin Ahn (Feb. 2025).

Academic Service

Conference/Journal Reviewer

International Conference on Learning Representations (ICLR) 2025–2027
 International Conference on Machine Learning (ICML) 2026 — **Silver Reviewer**
 Conference on Neural Information Processing Systems (NeurIPS) 2026
 Transactions on Machine Learning Research (TMLR) 2026–

Workshop Organizer

3rd KAIST-Mila Prefrontal AI Workshop (6 invited speakers incl. Yoshua Bengio), Mila, Aug. 2026, [Link](#)
 2nd KAIST-Mila Prefrontal AI Workshop (6 invited speakers incl. Yoshua Bengio), Mila, Jul. 2025, [Link](#)

Teaching Experience

- Spring 2026 **Teaching Assistant**, CS30707: Introduction to Reinforcement Learning, KAIST
- Fall 2025 **Teaching Assistant**, CS672: Reinforcement Learning, KAIST
- Spring 2025 **Teaching Assistant**, CS377: Introduction to Reinforcement Learning, KAIST
 - Outstanding Teaching Assistant Award recipient

Professional Experience

- Jun. 2021 – Jun. 2022 **MakinaRocks**, ML Research Engineer(Robotics Focused) Seoul, Korea
 - **Optimization of Welding Task and Motion Planning in Vehicle Manufacturing Process**
 - Challenges: 1. Large-scale and high complexity in Optimization Problem. 2. Required high-accuracy planning in highly cluttered environments. 3. Required tight Time Window for control
 - [Contribution 1] GA-based Multi-objective Optimization Module for Welding-point Assignment
 - [Contribution 2] Multi-Robot Interlock Scheduling System with Swept-volume-based collision detection
- Dec. 2017 – Jun. 2021 **Syscon, Co., Ltd.**, Robotics Software Engineer/Research Strategy Manager Incheon, Korea
 - **AMR Navigation System Design & Product Development**
 - [Field Exp.] **Successful Deployment for Smart Factory Project**
 - More than 300 delivery tasks per day with 14 robots
 - [Research Exp.] **Learning-based High-accuracy Object Pose Estimator(Dock Pose Estimation)**
 - [Research Exp.] **Distilled Meta Policy Navigation System for Multi-behavior Skillset and its Strategies**